01 Project Planning & Management

Global Crises & Their Economic Impact on Egypt | 2020 - October 2026

Project overview

This project examines how global crises coincided with movements in three selected market indicators: the official USD/EGP exchange rate, crude oil price, and global gold price. It then explores their relationship with annual headline inflation in Egypt using monthly observations.

Research question

How did major global crises from 2020 through October 2026 relate to movements in USD/EGP, oil, and gold, and what can those patterns tell us about inflationary pressures in Egypt?

Main objectives

- Build a reliable, documented monthly dataset and an interactive Power BI report.
- Compare movements in all three indicators across selected crisis windows.
- Examine inflation patterns, including possible time lags, without claiming causation from correlation.
- Communicate three clear findings and three evidence-aligned policy recommendations.

Scope and boundaries

Included	Excluded from detailed analysis
COVID-19, Russia-Ukraine war, Gaza/Red Sea disruptions, regional Iran-Israel escalations, and key Egypt FX events.	Separate models for tourism, GDP, exports, imports, shipping revenue, or household survey microdata.
USD/EGP, oil (USD/barrel), gold (USD/troy ounce), Egypt headline CPI year-on-year (%).	Daily event-day causality estimates, economic forecasts, or unverified claims about citizens.

Historical quantitative coverage: January 2020 - September 2026 (81 complete months). October 2026 events may appear as contextual notes only; incomplete monthly observations are excluded.

02 Timeline & Deliverables

A short delivery plan for the approved team.

Execution schedule - proposed

Phase	Timing	Output
Dataset validation and source review	17-19 Oct 2026	Validated workbook, source list, scope sign-off
Power Query and data model	Following documentation approval	Clean tables, date dimension, relationships
DAX and report development	Next development sprint	Measures and three dashboard pages
Analysis, QA and presentation	Final sprint before project submission	Insights, six-slide presentation, repository files

The detailed dates after 19 October are provisional and must be aligned with the program schedule once announced.

Final deliverables

- One Power BI (.pbix) report containing three interactive pages.
- A cleaned Excel workbook, source registry and crisis timeline.
- Documented Power Query steps, data model and DAX measures.
- A short PowerPoint presentation (approximately six slides).
- One GitHub repository with README and the three required documentation PDFs.

Success metrics

Metric	Target
Data completeness	81 aligned full months across four indicators
Usability	Three readable pages with working date/crisis controls
Reproducibility	Every chart traced to a documented series and calculation
Storytelling	Three defensible findings and three associated recommendations

03 Team Roles & Governance

Five members | clear functional ownership and joint review.

Responsibility assignment

Team member(s)	Primary responsibilities
Ahmed Hassan & Sama Ahmed	DAX measures; three Power BI dashboard pages; visual and filter testing.
Mohamed Rizk & Farid Hesham	Excel/Power Query validation and cleaning; date table, relationships and model QA.
Nouraldin Hazem	Visual theme, narrative/storytelling structure, six-slide presentation, integration of findings.

Shared responsibilities

- All team members review dataset interpretation, results, documentation and final demo.
- DAX/visual owners agree on table and measure naming with modeling owners.
- Narrative owner validates every written insight against final measures and charts.
- One team GitHub repository; members commit or submit updates through the agreed workflow.

Project management workflow

Backlog / To Do -> In Progress -> Review -> Approved. Use short team check-ins after model completion, measures completion, and final dashboard testing.

Repository organization

/data (clean source workbook), /powerbi (.pbix), /docs (three PDFs), /presentation (slides), and README.md (scope, tools, setup, sources).

Open administrative fields

GitHub Repository URL:

https://github.com/Ahmed-Bayam/Egypt-s-Resilience-The-Impact-of-Global-Crises-2020-2025-

Team leader / submission owner: Farid Hesham

04 Risk Management & Acceptance

Keep the project realistic, traceable and deliverable.

Risk register

Risk	Impact	Mitigation
Incomplete October 2026 values	Misleading comparisons	Use completed months through September 2026; label later events.
Overlapping global crises	Incorrect attribution	Identify event overlaps; report associations, not isolated causal effects.
Different units and frequencies	Invalid chart comparisons	Use common monthly dates; normalized index (=100) for comparative chart.
Model or DAX errors	Wrong KPIs	Check measures against Excel test values; peer-review filters and relationships.
Scope expansion	Missed deadline	Limit to four indicators, three dashboards, three conclusions.

Acceptance checklist

- Dataset sources and date boundaries are recorded.
- Month-level model joins are validated; no accidental many-to-many relationships.
- Dashboard 1: timeline and USD/EGP, oil, gold trends.
- Dashboard 2: inflation, comparisons, and indexed indicator views.
- Dashboard 3: three verified findings, three recommendations, and limitations.
- Documentation PDFs and project assets uploaded to one team GitHub repository.

Submission note

The documentation task specifies Monday, 19 October 2026. The team must register the exact project name and its single GitHub URL in the assigned Google Drive sheet, then submit according to LMS instructions.

01 Literature Review

Comparable public resources and how our project uses their lessons.

Purpose and search approach

We reviewed established macroeconomic data services and economic research rather than assuming that an identical student Power BI project already exists. The goal is to identify dependable sources, analytical strengths, and gaps our integrated dashboard can address.

Review selection

- World Bank Commodity Markets / Pink Sheet: official historical commodity price observations.
- Central Bank of Egypt (CBE) historical exchange-rate statistics: official USD/EGP observations.
- World Bank Egypt Economic Monitor: contextual discussion of Egypt economic trends.
- IMF 2025 Egypt Article IV / program materials: economic assessment and policy context.

Review criteria

Coverage, source credibility, accessibility, comparability, interpretation, user interaction, Egypt relevance, and whether different indicators appear in one coherent narrative.

Key observation

The selected resources are authoritative but designed for distinct purposes: commodity data, exchange-rate data, country monitoring, and policy analysis. Our project aims to present selected variables together in one accessible and interactive report.

This is a comparison with published resources, not a claim that these organizations offer the same Power BI project or identical visuals.

02 Existing Resources

Strengths and limitations relative to our specific use case.

Resource	Strengths	Limitation for our use
World Bank Pink Sheet	Consistent monthly oil and gold prices with known units.	Not an Egypt-specific crisis-to-inflation dashboard.
CBE exchange-rate history	Official historical USD/EGP series.	Needs integration with global commodities and a narrative.
Egypt Economic Monitor	Interpretation of Egypt developments, with economic context.	Periodic reports; not a single user-controlled monthly comparison.
IMF Egypt assessments	Macroeconomic indicators and policy discussion.	Written economic assessment, not our selected three-market interactive view.

Lessons learned

- Favor direct institutional datasets over unexplained third-party tables.
- Clearly disclose units: EGP/USD, USD/barrel, USD/troy ounce, CPI YoY %.
- Separate dated events and contextual commentary from measured price observations.
- Tell a cautious story: parallel movements do not establish a direct causal chain.

Why integration matters

A reader otherwise has to open several datasets and long reports to compare time windows. Our consolidated model provides synchronized dates, filters, indexed comparison, and a short Egypt-focused explanation.

03 Comparison & Project Gap

A practical improvement over consulting disconnected references.

Comparison matrix

Feature	Reference resources	Proposed DEPI project
USD/EGP series	CBE standalone data	Integrated monthly chart and filters
oil and gold series	World Bank commodity dataset	Both markets alongside USD/EGP
Egypt inflation context	Reports and inflation series	Direct comparison with monthly series
Global crisis markers	Explained in separate reports	One documented crisis timeline
Interactive exploration	Varies by resource	Three-page Power BI dashboard
Practical takeaways	Policy recommendations in reports	Three dataset-linked conclusions and recommendations

Distinctive contribution

The novelty is not creating new economic indicators. It is the curated comparison of widely used indicators over the same months, with documented crisis dates and Egypt-focused storytelling in a compact dashboard that a non-specialist can explore.

Methodological limitations

- Monthly averages can hide short-term market moves around event days.
- Crises overlap, and domestic exchange-rate policy can influence USD/EGP independently.
- Gold is a market/safe-haven indicator rather than a direct cause of CPI inflation.
- Inflation is a YoY percentage rate, not the absolute price level.

Therefore the final report will avoid statements such as "war X caused all inflation" without a causal research design.

04 Literature References

Official sources used to inform the review and methodology.

Verified public resources

[1] World Bank - Commodity Markets and Pink Sheet Data.
https://www.worldbank.org/en/research/commodity-markets

[2] Central Bank of Egypt - Exchange Rates Historical Data.
https://www.cbe.org.eg/en/economic-research/statistics/exchange-rates/historical-data

[3] World Bank - Egypt Economic Monitor.
https://www.worldbank.org/en/country/egypt/publication/egypt-economic-monitor

[4] IMF - Egypt 2025 Article IV Consultation and program review (March 2025).
https://www.imf.org/en/news/articles/2025/03/11/pr-2558-egypt-imf-completes-4th-rev-eff-arrangement-under-rsf-concl-2025-art-iv-consult

How each reference supports the project

Reference	Use
World Bank Commodity Markets	Commodity price data and conventions for oil/gold series.
CBE historical rates	Official exchange-rate source and validation.
Egypt Economic Monitor	Narrative background and interpretation of Egypt macro trends.
IMF Egypt assessment	Economic context, caveats, and policy discussion.

Research integrity

The attached project workbook remains the working dataset. The final report will keep source links, date coverage and aggregation definitions; any event claims should be checked against credible dated sources before publication.

Accessed for documentation preparation: October 2026.

01 Requirements Gathering

A focused definition of user needs and Power BI functionality.

Problem statement

Global crises, commodity markets, currency movements, and inflation are usually reviewed separately. Students and non-specialist readers need a single, accessible way to view changes during selected periods and understand how these series relate to Egypt.

Stakeholder groups

Stakeholder	Need
DEPI instructors/evaluators	Clear scope, sound modeling, reproducible metrics, documented sources.
Students and general readers	Simple timelines, explanations, filters, understandable charts.
Team analysts/developers	Stable schemas, reusable measures, consistent monthly dates.
Team presenter	A concise evidence-led narrative and exportable visuals.

User needs / user stories

- As a reader, I want to select a crisis so I can inspect indicator behavior around it.
- As an analyst, I want synchronized monthly series so I can compare trends fairly.
- As an evaluator, I want visible sources, units, definitions, and limits.
- As a presenter, I want three major conclusions explained in plain language.

Scope control

Only USD/EGP, oil, global gold and Egypt headline CPI are modeled quantitatively. Other themes - imports, Suez Canal, tourism - provide brief context only.

02 Functional Requirements

What the finished Power BI report must provide.

ID	Requirement	Priority
FR-01	Import four monthly time-series tables and documented crisis events.	Must
FR-02	Create a shared monthly calendar model for consistent date filtering.	Must
FR-03	Show USD/EGP, oil and gold trend charts with units.	Must
FR-04	Filter by month/year and selected crisis comparison window.	Must
FR-05	Show headline CPI inflation trend for Egypt.	Must
FR-06	Calculate month-to-month and selected-period percentage changes.	Must
FR-07	Provide indexed comparison (common base of 100).	Should
FR-08	Present three checked findings and recommendations.	Must
FR-09	Display sources/method notes and incomplete-month warning.	Must
FR-10	Enable export of figures / integration in presentation.	Should

Dashboard structure

- Page 1 - Global Crises Timeline: crisis context plus USD/EGP, oil and gold trends.
- Page 2 - Economic Impact on Egypt: CPI inflation and comparative market movements.
- Page 3 - Findings & Solutions: three key results, policy options, limitations.

Explicit exclusions

No live stock/commodity trading, forecasts, personal accounts, user data collection, or direct measurement of individual households.

03 Data & Technical Requirements

Validated monthly source tables, model relationships and calculations.

Source tables

Table	Essential fields
USD-EGP	Date, USD EGP Monthly Avg Sell, Status, Source
oil	Date, oil USD per bbl, Status, Source
Gold	Date, Gold USD per troy oz, Status, Source
Egypt Inflation	Date, Headline CPI YoY Pct, Status, Basis, Source
Crisis Timeline	Event ID, Event Name, Start Date, End Date, event context
Source Registry	Dataset metadata and source links; documentation only

Transformation and model requirements

- Use one observation per completed month per indicator; exclude partial October 2026.
- Enforce Date and Decimal Number types and review blank/duplicate keys.
- Build Dim_Date for common monthly filtering (one-to-many where appropriate).
- Keep event periods separate or implement comparison logic that safely handles overlapping events.
- Retain inflation values as percent points where, e.g., 13.9 means 13.9%, not 1390%.

Measure requirements

Monthly indicator values; MoM percentage change; selected-period change; indexed base-100 series; inflation values; optional lag analysis only if supported and correctly labeled.

Tool stack

Microsoft Excel, Power Query, Power BI Desktop, DAX, PowerPoint, and GitHub.

04 Non-Functional Requirements & Tests

The finished solution should be clear, reliable and easy to review.

Non-functional requirements

Aspect	Requirement
Accuracy	Source-linked values, visible units, documented calculations and month coverage.
Performance	Report interaction should be responsive with the small monthly dataset.
Usability	Consistent layout, readable legends, descriptive titles and simple slicers.
Accessibility	Contrast and labels not reliant on color alone.
Security	Only public economic data; no credentials or personal information in repository.
Maintainability	Readable names, organized measures and documented data refresh steps.
Reproducibility	Another team member can regenerate results from the workbook and PBIX.

Acceptance tests

- All four series align from Jan 2020 to Sep 2026 in the core comparison.
- Partial-month rows are excluded from headline quantitative comparisons.
- Date filter changes each relevant visual as expected.
- Crisis selection does not unintentionally eliminate unrelated data or create invalid joins.
- Displayed numbers match checked records and DAX calculations.
- Charts disclose index base, percentage units and relevant limitations.
- Three-page dashboard and six-slide presentation tell one consistent story.

Sign-off

Review owners: Mohamed Rizk & Farid Hesham (data/model); Ahmed Hassan & Sama Ahmed (DAX/dashboard); Nouraldin Hazem (theme/story/presentation). Final approval: all five team members.

Repository URL: https://github.com/Ahmed-Bayam/Egypt-s-Resilience-The-Impact-of-Global-Crises-2020-2025-